

LECTURE 4

CONFIDENCE INTERVALS

In this lecture, we consider the normal regression model defined by Assumptions (A1)-(A5).

The *point estimator* $\hat{\beta}$ of the vector of parameters β is not very informative, since $\Pr(\hat{\beta} = \beta) = 0$. We consider the construction of *random* intervals or regions with the property that they include the true parameter with some specified probability $1 - \alpha$, where α is a small number ($1 - \alpha$ is called the *confidence level*). Usually, the following values of α are considered: 0.01, 0.05, and 0.10. A confidence interval with *coverage probability* $1 - \alpha$ is denoted by $CI_{1-\alpha}$.

Scalar case

Suppose that we are interested in constructing a confidence interval for β_1 , the first element of β . The partitioned regression is given by

$$Y = \beta_1 X_1 + X_2 \beta_2 + U,$$

where X_1 is the $n \times 1$ vector containing all n observations on the first regressor. The OLS estimator of β_1 is

$$\hat{\beta}_1 = \frac{X_1^\top M_2 Y}{X_1^\top M_2 X_1},$$

where $M_2 = I_n - X_2 (X_2^\top X_2)^{-1} X_2^\top$.

One method of constructing confidence intervals is to consider *symmetric* intervals around the point estimator:

$$CI_{1-\alpha} = [\hat{\beta}_1 - c, \hat{\beta}_1 + c]. \quad (1)$$

Since $\hat{\beta}_1$ is a function of the random sample, the confidence interval $CI_{1-\alpha}$ as defined in (1) is random as well. The problem is to choose c so that

$$\Pr(\beta_1 \in CI_{1-\alpha} \mid X) = 1 - \alpha,$$

where $X = \begin{pmatrix} X_1 & X_2 \end{pmatrix}$. To find c , one must know the conditional distribution of $\hat{\beta}_1$ given X . Under Assumptions (A1)-(A5),

$$\begin{aligned} \hat{\beta}_1 \mid X &\sim N(\beta_1, \sigma^2 / (X_1^\top M_2 X_1)), \text{ and, consequently,} \\ \frac{\hat{\beta}_1 - \beta_1}{\sqrt{\sigma^2 / (X_1^\top M_2 X_1)}} \mid X &\sim N(0, 1). \end{aligned} \quad (2)$$

To verify this, note that $\hat{\beta}_1$ is a linear estimator, and write $\hat{\beta}_1 = \beta_1 + (X_1^\top M_2 U) / (X_1^\top M_2 X_1)$.

Let z_τ be the τ -quantile of the standard normal distribution; that is, if $Z \sim N(0, 1)$, then

$$\Pr(Z \leq z_\tau) = \tau.$$

For example, for $\tau = 0.5$ we have the *median*:

$$\Pr(Z \leq z_{0.5}) = 0.5.$$

Since the standard normal distribution is symmetric around zero,

$$\begin{aligned} z_\alpha &= -z_{1-\alpha}, \text{ and, therefore,} \\ \Pr(-z_{1-\alpha/2} \leq Z \leq z_{1-\alpha/2}) &= 1 - \alpha. \end{aligned}$$

For example, for $\alpha = 0.05$, $z_{1-0.05/2} = z_{0.975} = 1.96$, and $z_{0.025} = -1.96$.

σ^2 is known

Suppose for a moment that σ^2 is known, so that the variance of $\hat{\beta}_1$ can be computed exactly. We set

$$\begin{aligned} c &= z_{1-\alpha/2} \sqrt{\text{Var}(\hat{\beta}_1 | X)} \\ &= z_{1-\alpha/2} \sqrt{\sigma^2 / (X_1^\top M_2 X_1)}. \end{aligned}$$

We will show that

$$\Pr \left(\beta_1 \in \left[\hat{\beta}_1 - z_{1-\alpha/2} \sqrt{\sigma^2 / (X_1^\top M_2 X_1)}, \hat{\beta}_1 + z_{1-\alpha/2} \sqrt{\sigma^2 / (X_1^\top M_2 X_1)} \right] \mid X \right) = 1 - \alpha.$$

Indeed,

$$\begin{aligned} & \Pr \left(\hat{\beta}_1 - z_{1-\alpha/2} \sqrt{\sigma^2 / (X_1^\top M_2 X_1)} \leq \beta_1 \leq \hat{\beta}_1 + z_{1-\alpha/2} \sqrt{\sigma^2 / (X_1^\top M_2 X_1)} \mid X \right) \\ &= \Pr \left(-z_{1-\alpha/2} \sqrt{\sigma^2 / (X_1^\top M_2 X_1)} \leq \beta_1 - \hat{\beta}_1 \leq z_{1-\alpha/2} \sqrt{\sigma^2 / (X_1^\top M_2 X_1)} \mid X \right) \\ &= \Pr \left(-z_{1-\alpha/2} \sqrt{\sigma^2 / (X_1^\top M_2 X_1)} \leq \hat{\beta}_1 - \beta_1 \leq z_{1-\alpha/2} \sqrt{\sigma^2 / (X_1^\top M_2 X_1)} \mid X \right) \\ &= \Pr \left(-z_{1-\alpha/2} \leq \frac{\hat{\beta}_1 - \beta_1}{\sqrt{\sigma^2 / (X_1^\top M_2 X_1)}} \leq z_{1-\alpha/2} \mid X \right). \end{aligned} \tag{3}$$

The desired result follows immediately from (2), (3), and the definition of $z_{1-\alpha/2}$.

σ^2 is unknown

The construction of $CI_{1-\alpha}$ above relies on σ^2 being known. When σ^2 is unknown, one can take a similar approach by first replacing σ^2 with an estimator, for example:

$$s^2 = \hat{U}^\top \hat{U} / (n - k).$$

However, $(\hat{\beta}_1 - \beta_1) / \sqrt{s^2 / (X_1^\top M_2 X_1)}$ is not normally distributed, since it is a nonlinear function of the random $\hat{\beta}_1$ and s^2 . Therefore, one cannot use the quantiles of the standard normal distribution for the construction of confidence intervals.

The key result is that

$$\frac{(\hat{\beta}_1 - \beta_1)}{\sqrt{s^2 / (X_1^\top M_2 X_1)}} \mid X \sim t_{n-k}. \tag{4}$$

Recall that the t_{n-k} distribution is defined as the distribution of

$$Z / \sqrt{V / (n - k)},$$

where Z is a standard normal random variable, V is a χ_{n-k}^2 random variable, and Z and V are independent. Write

$$\begin{aligned} \frac{\hat{\beta}_1 - \beta_1}{\sqrt{s^2 / (X_1^\top M_2 X_1)}} &= \left(\frac{\hat{\beta}_1 - \beta_1}{\sqrt{\sigma^2 / (X_1^\top M_2 X_1)}} \right) / \sqrt{\frac{s^2}{\sigma^2}} \\ &= \left(\frac{\hat{\beta}_1 - \beta_1}{\sqrt{\sigma^2 / (X_1^\top M_2 X_1)}} \right) / \sqrt{\frac{\hat{U}^\top \hat{U}}{\sigma^2} / (n - k)}. \end{aligned} \tag{5}$$

We already know that, in the above expression, $(\hat{\beta}_1 - \beta_1) / \sqrt{\sigma^2 / (X_1^\top M_2 X_1)} \mid X \sim N(0, 1)$. We show next that, conditional on X ,

$$\frac{\hat{U}^\top \hat{U}}{\sigma^2} \mid X \sim \chi_{n-k}^2. \quad (6)$$

To show this, we need the following result:

Lemma 1. *Suppose that the n -vector $U \sim N(0, I_n)$. Let A be a non-random $n \times n$ symmetric and idempotent matrix with $\text{rank}(A) = r \leq n$. Then, $U^\top A U \sim \chi_r^2$.*

Proof. It suffices to show that $U^\top A U = \sum_{i=1}^r Z_i^2$, where Z_i are iid $N(0, 1)$.

Since A is a symmetric matrix, one can write

$$A = C \Lambda C^\top,$$

where Λ is an $n \times n$ diagonal matrix consisting of the eigenvalues of A , and $C^\top C = I_n$. Since A is idempotent,

$$\begin{aligned} A &= A A, \text{ and} \\ C \Lambda C^\top &= (C \Lambda C^\top) (C \Lambda C^\top) \\ &= C \Lambda^2 C^\top. \end{aligned}$$

Therefore,

$$\Lambda = \Lambda^2,$$

which implies that each eigenvalue of A is either zero or one. Since the rank of a matrix equals the number of its non-zero eigenvalues, there are r non-zero eigenvalues λ_i in

$$\Lambda = \begin{pmatrix} \lambda_1 & 0 & \dots & 0 \\ 0 & \lambda_2 & \dots & 0 \\ \dots & \dots & \dots & \dots \\ 0 & 0 & \dots & \lambda_n \end{pmatrix}.$$

Define

$$Z = C^\top U.$$

Since $U \sim N(0, I_n)$, we have that

$$\begin{aligned} Z &\sim N(0, C^\top C) \\ &\sim N(0, I_n). \end{aligned}$$

Lastly,

$$\begin{aligned} U^\top A U &= Z^\top \Lambda Z \\ &= \sum_{i=1}^n \lambda_i Z_i^2. \end{aligned}$$

The result follows because Z_i are iid $N(0, 1)$, there are r eigenvalues equal to one, and the remaining $n - r$ eigenvalues are zero. $\square$

Now, to show (6), write

$$\frac{\hat{U}^\top \hat{U}}{\sigma^2} = \left(\frac{U}{\sigma} \right)^\top M_X \left(\frac{U}{\sigma} \right), \quad (7)$$

where

$$M_X = I_n - X (X^\top X)^{-1} X^\top.$$

By Assumption (A5),

$$\frac{U}{\sigma} \mid X \sim N(0, I_n). \quad (8)$$

Since M_X is symmetric and idempotent, its eigenvalues are either zero or one. Therefore,

$$\begin{aligned} \text{rank}(M_X) &= \text{tr}(M_X) \\ &= n - k. \end{aligned} \quad (9)$$

The result in (6) follows from (7), (8), (9), and Lemma 1.

Finally, we show that $\hat{\beta}_1 - \beta_1$ and $\hat{U}^\top \hat{U}$ in (5) are independent given X . Write

$$\begin{aligned} \hat{\beta}_1 - \beta_1 &= (X_1^\top M_2 U) / (X_1^\top M_2 X_1), \\ \hat{U}^\top \hat{U} &= U^\top M_X U. \end{aligned}$$

It is sufficient to show independence of $X_1^\top M_2 U$ and $M_X U$. Since $\hat{\beta}_1$ is a function of $X_1^\top M_2 U$, and $\hat{U}^\top \hat{U}$ is a function of $M_X U$, independence of $X_1^\top M_2 U$ and $M_X U$ implies independence of $\hat{\beta}_1$ and $\hat{U}^\top \hat{U}$. First, we show that they are uncorrelated:

$$\begin{aligned} \text{Cov}(X_1^\top M_2 U, M_X U \mid X) &= E[X_1^\top M_2 U U^\top M_X \mid X] \\ &= X_1^\top M_2 E[U U^\top \mid X] M_X \\ &= X_1^\top M_2 (\sigma^2 I_n) M_X \\ &= \sigma^2 X_1^\top M_2 M_X \\ &= \sigma^2 X_1^\top M_X \text{ (see Lecture 3, page 8)} \\ &= 0. \end{aligned}$$

Since $X_1^\top M_2 U$ and $M_X U$ are both linear functions of U , they are jointly normal conditional on X . Joint normality and zero correlation imply independence. Consequently, $\hat{\beta}_1 - \beta_1$, a function of $X_1^\top M_2 U$, and $\hat{U}^\top \hat{U}$ are independent as well.

We have shown (4). Consequently, when constructing confidence intervals, if one replaces the unknown σ^2 with s^2 , one must replace $z_{1-\alpha/2}$ with quantiles of the t distribution, $t_{n-k, 1-\alpha/2}$:

$$CI_{1-\alpha} = \left[\hat{\beta}_1 - t_{n-k, 1-\alpha/2} \sqrt{s^2 / (X_1^\top M_2 X_1)}, \hat{\beta}_1 + t_{n-k, 1-\alpha/2} \sqrt{s^2 / (X_1^\top M_2 X_1)} \right].$$

The expression $s^2 / (X_1^\top M_2 X_1)$ that appears in the above equation is the *estimated* variance of $\hat{\beta}_1$:

$$\text{Var}(\hat{\beta}_1 \mid X) = s^2 / (X_1^\top M_2 X_1).$$

Thus, one constructs a $(1 - \alpha)$ -level confidence interval for β_j , $j = 1, \dots, k$, as follows:

$$CI_{1-\alpha}^j = \left[\hat{\beta}_j - t_{n-k, 1-\alpha/2} \sqrt{\text{Var}(\hat{\beta}_j \mid X)}, \hat{\beta}_j + t_{n-k, 1-\alpha/2} \sqrt{\text{Var}(\hat{\beta}_j \mid X)} \right]. \quad (10)$$

Vector case

Suppose that we are concerned with the vector of parameters $\beta = (\beta_1, \beta_2, \dots, \beta_k)$. Equation (10) describes how to construct individual confidence intervals for the elements of β . These are concerned with *marginal* distributions of the elements of β , and their simple combination does not produce a set that includes the whole vector β with a desired probability. In this section, we construct random *regions* that contain β with a specified probability $1 - \alpha$. We keep the notation $CI_{1-\alpha}$, even though in this case $CI_{1-\alpha}$ is a subset of $\mathbb{R}^k$.

The following is a simple and conventional approach to constructing confidence regions. We are looking for a *random* region $CI_{1-\alpha} \subseteq \mathbb{R}^k$ such that $\Pr(\beta \in CI_{1-\alpha} | X) = 1 - \alpha$. Consider a quadratic form in $(\hat{\beta} - \beta)$:

$$\begin{aligned} & (\hat{\beta} - \beta)^\top \left(\widehat{\text{Var}}(\hat{\beta} | X) \right)^{-1} (\hat{\beta} - \beta) / k \\ &= (\hat{\beta} - \beta)^\top \left(s^2 (X^\top X)^{-1} \right)^{-1} (\hat{\beta} - \beta) / k \\ &= \frac{(\hat{\beta} - \beta)^\top \left(\sigma^2 (X^\top X)^{-1} \right)^{-1} (\hat{\beta} - \beta) / k}{s^2 / \sigma^2} \\ &= \frac{(\hat{\beta} - \beta)^\top \left(\sigma^2 (X^\top X)^{-1} \right)^{-1} (\hat{\beta} - \beta) / k}{\left(\frac{\hat{U}^\top \hat{U}}{\sigma^2} \right) / (n - k)}. \end{aligned} \quad (11)$$

Next, we show that the expression in (11) has the $F_{k, n-k}$ distribution given X .

The $F_{k, n-k}$ distribution is defined as the distribution of

$$\frac{V/k}{W/(n-k)},$$

where $V \sim \chi_k^2$, $W \sim \chi_{n-k}^2$, and V and W are independent. From the discussion in the previous section, we know that $\hat{U}^\top \hat{U} / \sigma^2 | X \sim \chi_{n-k}^2$ and that it is independent of the numerator in (11). Thus, we need to show that

$$(\hat{\beta} - \beta)^\top \left(\sigma^2 (X^\top X)^{-1} \right)^{-1} (\hat{\beta} - \beta) | X \sim \chi_k^2. \quad (12)$$

We need the following lemma.

Lemma 2. Suppose that the k -vector $U \sim N(0, \Sigma)$, where Σ is a positive definite variance-covariance matrix. Then, $U^\top \Sigma^{-1} U \sim \chi_k^2$.

Proof. Since Σ is symmetric, $\Sigma = C\Lambda C^\top$, where Λ is a diagonal matrix of the eigenvalues of Σ on its main diagonal, and $C^\top C = CC^\top = I_k$. Since Σ is positive definite, its eigenvalues are positive, and therefore, $\Lambda^{1/2}$ can be defined as

$$\Lambda^{1/2} = \begin{pmatrix} \lambda_1^{1/2} & & 0 \\ & \ddots & \\ 0 & & \lambda_k^{1/2} \end{pmatrix},$$

and $\Lambda^{-1/2}$ can be defined as

$$\Lambda^{-1/2} = \begin{pmatrix} \lambda_1^{-1/2} & & 0 \\ & \ddots & \\ 0 & & \lambda_k^{-1/2} \end{pmatrix}.$$

Next, since $C\Lambda^{-1}C^\top C\Lambda C^\top = I_k$, we have that

$$\Sigma^{-1} = C\Lambda^{-1}C^\top.$$

Now, define

$$\Sigma^{1/2} = C\Lambda^{1/2}C^\top \text{ and } \Sigma^{-1/2} = C\Lambda^{-1/2}C^\top.$$

We have that $(\Sigma^{1/2})^\top = \Sigma^{1/2}$ and $(\Sigma^{-1/2})^\top = \Sigma^{-1/2}$ (symmetric). Furthermore,

$$\begin{aligned} \Sigma^{1/2}\Sigma^{1/2} &= C\Lambda^{1/2}C^\top C\Lambda^{1/2}C^\top = C\Lambda^{1/2}\Lambda^{1/2}C^\top = C\Lambda C^\top = \Sigma, \\ \Sigma^{-1/2}\Sigma\Sigma^{-1/2} &= C\Lambda^{-1/2}C^\top C\Lambda C^\top C\Lambda^{-1/2}C^\top = C\Lambda^{-1/2}\Lambda\Lambda^{-1/2}C^\top = CC^\top = I_k. \end{aligned}$$

The matrix $\Sigma^{1/2}$ is called the symmetric square root of Σ , and $\Sigma^{-1/2}$ is the inverse symmetric square root. Define a k -vector

$$V = \Sigma^{-1/2}U,$$

so that

$$U^\top \Sigma^{-1}U = V^\top V. \quad (13)$$

Since $U \sim N(0, \Sigma)$, and V is a linear transformation of U , we have that

$$\begin{aligned} V &\sim N\left(0, \Sigma^{-1/2} \text{Var}(U) \Sigma^{-1/2}\right) \\ &= N\left(0, \Sigma^{-1/2} \Sigma \Sigma^{-1/2}\right) \\ &= N(0, I_k). \end{aligned}$$

From (13) and the definition of the χ_k^2 distribution,

$$U^\top \Sigma^{-1}U = V^\top V = \sum_{j=1}^k V_j^2 \sim \chi_k^2. \quad \square$$

Now, the result in (12) follows from Lemma 2. Consequently,

$$\frac{\left(\widehat{\beta} - \beta\right)^\top \left(s^2 (X^\top X)^{-1}\right)^{-1} \left(\widehat{\beta} - \beta\right)}{k} \mid X \sim F_{k, n-k}.$$

Let $F_{k, n-k, \tau}$ be the τ -quantile of the $F_{k, n-k}$ distribution. We construct the $(1-\alpha)$ -level confidence region as follows:

$$CI_{1-\alpha} = \left\{ b \in \mathbb{R}^k : \left(\widehat{\beta} - b\right)^\top \left(s^2 (X^\top X)^{-1}\right)^{-1} \left(\widehat{\beta} - b\right) / k \leq F_{k, n-k, 1-\alpha} \right\}.$$

From the above discussion, it follows that

$$\begin{aligned} &\Pr(\beta \in CI_{1-\alpha} \mid X) \\ &= \Pr\left(\left(\widehat{\beta} - \beta\right)^\top \left(s^2 (X^\top X)^{-1}\right)^{-1} \left(\widehat{\beta} - \beta\right) / k \leq F_{k, n-k, 1-\alpha} \mid X\right) \\ &= 1 - \alpha. \end{aligned}$$

Remark

The confidence interval/region $CI_{1-\alpha}$ is a function of the sample $\{(Y_i, X_i) : i = 1, \dots, n\}$, and is therefore random, which allows us to talk about the probability of $CI_{1-\alpha}$ containing the true value of β . On the other hand, the realization of $CI_{1-\alpha}$ is not random. Once the confidence interval is computed given the data, it does not make sense anymore to talk about the probability that it includes β . It is either zero or one.